

Extra Question.pdf

Exported: May 28, 2026

Summary

This document provides a series of practical exercises designed to master **NoSQL database management** using Apache Cassandra. Through various scenarios like student databases and e-commerce systems, it guides learners through **fundamental CRUD operations** and the structural organization of data using keyspaces and tables. The curriculum specifically highlights advanced features such as **efficient batch processing**, the manipulation of **collection data types**, and the implementation of **temporary data storage** through Time To Live (TTL) settings. Ultimately, these tasks serve as a structured roadmap for understanding how to **dynamically alter schemas** and manage real-world data lifecycles in a distributed environment.

Keywords: Cassandra database operations, CRUD operations, Collection data types, Time To Live, Table schema manipulation

Content

Practice Question 1: Student Database

Perform the following operations using Cassandra:

- a) Create a keyspace named College
- b) Create a table Student_Info with attributes:
USN (Primary Key), Name, Branch, Semester, CGPA
- c) Insert at least 3 student records
- d) Update the CGPA of a student with a given USN
- e) Display all student records

Focus: Basic CRUD operations

Practice Question 2: Book Management System

Perform the following operations using Cassandra:

- a) Create a keyspace named Library
- b) Create a table Book_Info with attributes:
Book_Id (Primary Key), Title, Author, Price, Category

- c) Insert multiple records using BATCH
- d) Update the price of a specific book
- e) Delete a book record using Book_Id

Focus: BATCH, UPDATE, DELETE

Practice Question 3: Customer Orders

Perform the following operations using Cassandra:

- a) Create a keyspace named Ecommerce
- b) Create a table Customer_Orders with attributes:
Order_Id (Primary Key), Customer_Name, Product_Name, Quantity,
Order_Date
- c) Insert at least 3 order records
- d) Add a new column Delivery_Status using ALTER
- e) Update the delivery status for a given order

Focus: ALTER TABLE + UPDATE

Practice Question 4: Employee Skills (Collection Data Type)

Perform the following operations using Cassandra:

- a) Create a keyspace named Company
- b) Create a table Employee_Skills with attributes:
Emp_Id (Primary Key), Emp_Name, Skills (SET)
- c) Insert employee records with multiple skills
- d) Add a new skill to an existing employee
- e) Display employee names along with skills

Focus: SET collection data type

Practice Question 5: Product Expiry using TTL

Perform the following operations using Cassandra:

- a) Create a keyspace named Store
- b) Create a table Product_Info with attributes:
Product_Id (Primary Key), Product_Name, Price, Category
- c) Insert a product record with TTL of 20 seconds
- d) Display the product immediately
- e) Display the product after 20 seconds and observe the result



Focus: TTL (Time To Live)